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A Refined Mud Gas Ratio Method for Fluid Identification While Drilling

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Abstract

Mud logging, a cost-effective surface measurement, offers a real-time and high-resolution mud gas data for monitoring drilling operations and identifying pay zones and trapped fluids in subsurface formations. However, traditional fluid typing methods of mud gas, such as Haworth ratios and Pixler cross-plot, yield inconsistent results with established petroleum engineering classification. This paper presents a refined approach that builds upon the traditional Haworth method, incorporating three key parameters: gas wetness (Wh), the differential between Wh and balance ratio (Bh), and the methane-to-total-hydrocarbon-gas ratio (C1R). This enhanced methodology enables the identification of eight distinct hydrocarbon fluid types and has been validated through a computer program that automates data quality checks, log visualization, and fluid type determination. The software has been tested on over 100 wells in a basin-scale study, establishing a set of cutoffs for fluid typing in the Paleozoic petroleum system. The predicted fluids agree with fluid samples, source rock maturity, and basin modeling predictions. Three well studies are selected to demonstrate the effectiveness of this approach in generating high-resolution fluid profiles, identifying fluids for multiple pay zones, and distinguishing between different gas types. This refined methodology establishes a robust link between mud gas ratios and fluid types, facilitating real-time fluid identification and supporting informed decisions regarding fluid and rock sampling, as well as optimizing well testing.

Introduction

Mud logging involves measuring gases released from drilling cuttings and circulated drilling fluid at the surface, providing a real-time and high-resolution data asset for identifying hydrocarbon-bearing zones and trapped fluids within subsurface formations (Dria, 2007; Bateman, 2012). As drilling progresses, the hydrocarbon gas (C1 - C5) released from the drilling mud and cuttings at the surface reflects the composition of the hydrocarbon fluid in subsurface formations. Gas chromatography (GC) connected with flame ionization detector (FID) is deployed in the field to measure concentrations (ppm) of hydrocarbon species in mud gas, providing a continuous record of gas readings, which is a critical component of mud log for monitoring drilling operations and formation evaluation.

Mud logging is a direct measurement of subsurface rock and fluid, yielding the first available and high-resolution well data for operators (Whittaker, 1991). As a drilling synchronous operation, mud logging

does not incur additional rig time, making it a cost-effective service compared to wireline logging and formation testing tools. However, mud gas data are currently underutilized in formation evaluation due to concerns over their reliability and representativeness of the formation fluids (Blanc et al., 2003; Kandel et al., 2001). Various factors, including changes in drilling operations, drill-bit metamorphism (DBM) (Wenger et al., 2009), mud type (oil-based vs. water-based mud), and instrumental shift, may introduce errors and uncertainties into the data, leading to inaccurate determination of hydrocarbon shows and fluid types when using raw mud gas data. To address these issues, new instrumentation, field calibration, extractor efficiency correction, quality control (QC) parameters and procedure have been developed to minimize noise and ensure standard and high-quality data for formation evaluation (Kandel et al., 2001; Qubaisi et al., 2024). As the development of QC techniques, standard mud gas (SMG) logging can provide reliable total hydrocarbon gas readings as hydrocarbon shows for pay zone identification; while advanced mud gas logging (AMG) is approaching the accurate formation gas composition (McKinney et al., 2007; Hekimoglu et al., 2018).

Mud gas data is also widely used for formation fluid typing and predicting fluid properties. Fluid typing involves identifying and classifying different fluid types and their boundaries within target formations. The composition of mud gas is indicative of fluid type and can be correlated with fluid properties, as illustrated in Fig. 1. Multiple mud gas ratios have been employed to represent mud gas composition, identify fluid types, and infer fluid properties while drilling (Pixler, 1969; Haworth et al., 1985; Pinna and Law, 2008). Recently, machine learning models have been developed to elucidate the complex relationship between mud gas, fluid composition, and PVT data, enabling successful prediction of certain fluid properties (such as GOR and API) for both unconventional and conventional reservoirs (Anifowose et al., 2022; Arief and Yang, 2024; Yang et al., 2024). Nevertheless, the mud gas ratio method remains the most commonly used and rapid approach for fluid typing. This paper focuses on enhancing the empirical method of mud gas ratios for fluid typing and introduces the automated workflow and log visualization using an in-house developed computer program.

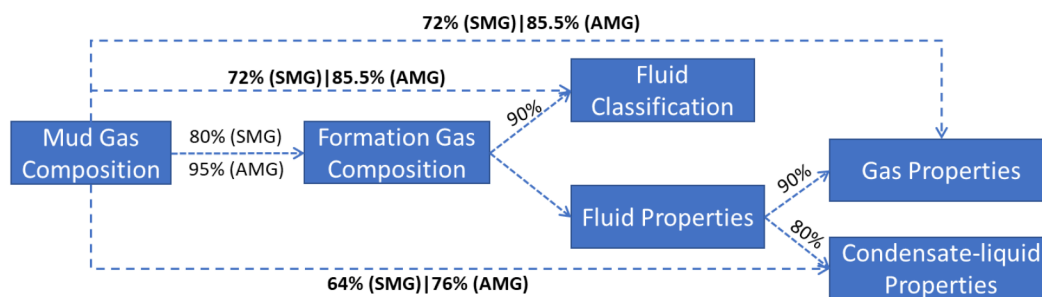


Figure 1—The correlation and assumed accuracy (%) from mud gas composition to fluid classification and properties (SMG: standard mud gas; AMG: advanced mud gas logging)

Workflow and software

We have developed and automated an integrated workflow that utilizes mud log data, including drilling parameters, mud gas and cutting lithology, to identify pay zones and fluids (Fig. 2). This workflow and the visualization of logs and interpretation are implemented in our in-house software, GeochemLog. To ensure data quality, multiple published and innovative QC parameters are used to evaluate the reliability of mud gas data, excluding poor-quality data in formation evaluation and leading to corresponding correction procedures. Two normalization methods of total hydrocarbon gas (TGas and THA) and DBM indicator based on alkene gases are developed to mitigate the impact of drilling operations on mud gas readings and enhance the identification of hydrocarbon show (Li and Luo, 2025). This paper presents our improvements of mud gas ratio method for fluid typing and showcases three application examples from well studies. Our goal is to reduce uncertainties and improve the accuracy of mud gas-fluid correlation (Fig. 1), as well as develop a standardized tool and streamline formation evaluation based on mud log.

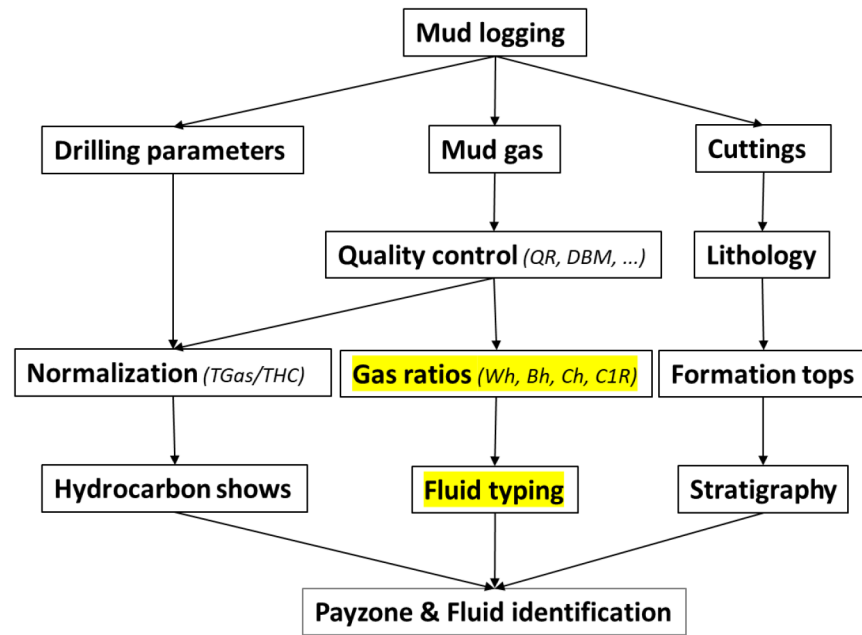


Figure 2—A formation evaluation workflow based on mud log data in GeochemLog software. The paper focuses on the improvement of mud gas ratio method for fluid typing

Mud Gas Ratio Methods

There are three commonly used mud gas ratio methods for fluid typing based on mud gas data.

Haworth (Whittaker) Ratios

The method uses three gas ratio parameters: gas wetness (Wh), balance ratio (Bh), and character ratio (Ch)] to determine fluid type (Haworth et al., 1985; Whittaker, 1991):

$$\text{Gas Wetness (Wh)} = \frac{C2 + C3 + iC4 + nC4 + iC5 + nC5}{C1 + C2 + C3 + iC4 + nC4 + iC5 + nC5} \quad (\text{Eq. 1})$$

$$\text{Balance Ratio (Bh)} = \frac{C1 + C2}{C3 + iC4 + nC4 + iC5 + nC5} \quad (\text{Eq. 2})$$

$$\text{Character Ratio (Ch)} = \frac{iC4 + nC4 + iC5 + nC5}{C3} \quad (\text{Eq. 3})$$

Fig. 3a illustrates an ideal log, along with the gas ratio cutoffs and fluid type determination. The Haworth (Whittaker) Ratios method would be the most common approach in the industry for fluid typing. The original method, as well as subsequent guidelines published by the Society of Petroleum Engineers (SPE) (Dara, 2007; Bateman, 2012) categorize fluids into six distinct types: dry gas, light/medium density gas, light oil, medium gravity oil, residual oil, and coal bed gas. However, there are some discrepancies between this fluid typing and the fluid classification used in reservoir and production engineering (Fig. 4). Furthermore, the qualitative comparison (Bh vs. Wh) described in Haworth et al. (1985) and Dara (2007), lacks the precision for coding purposes. Additionally, the direct use of LHR (Bh) and OCQ (Ch) cutoffs, as recommended by Bateman (2012) are not mathematically self-consistent, often resulting in inconclusive fluid types in our practice.

Pixler Ratios and Cross Plot

Pixler (1969) developed a method that was later refined by Ferrie (1981) to utilize a set of gas ratios, including C1/C2, C1/C3, C1/C4, and C1/C5, in conjunction with a cross-plot (Fig. 3b) to differentiate oi and gas and indicate the production potential. The Pixler method is often employed in tandem with Haworth ratios to provide a more comprehensive fluid typing. However, the method's reliance on four parameters

with subjective boundaries between oil, gas and nonproductive zones in the cross-plot poses challenges for full validation in computer program and log display. Furthermore, very dry gas has been produced from sandstone reservoir (without hydraulic fracturing) and unconventional shale reservoir, suggesting that the method need to be revised. The production of very dry gas from sandstone reservoirs and unconventional shale reservoirs suggests that the method may require revision.

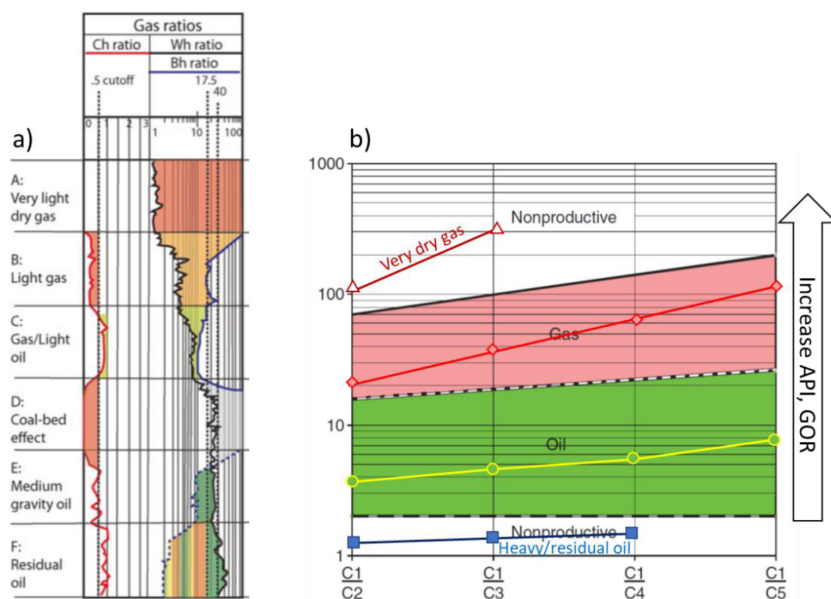


Figure 3—Two commonly used mud gas ratio methods for fluid typing: a) Haworth (Whittaker) ratios, from Dolson (2016); b) Pixler cross-plot, modified after Bateman (2012).

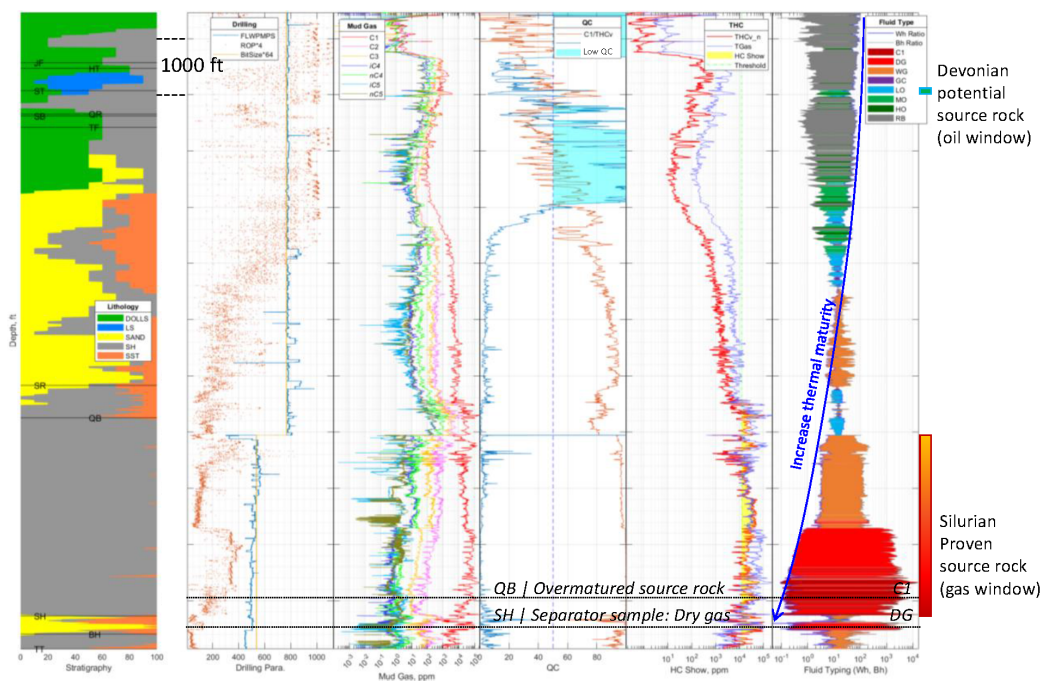


Figure 4—Well A: mud gas interpretation in GeochemLog demonstrates the evolution of fluid from heavy oil to very dry gas as increased depth in the Paleozoic petroleum system and extensive hydrocarbon shows in SR, QB and SH formations. The width of Wh and Bh curves represent the parameter "Wh-Bh". Dashed lines indicate the depths of rock/fluid samples. See fluid ID in Table 1.

Table 1—The comparison of Haworth ratio method to GeochemLog method for fluid typing and the general relation between fluid properties and mud gas ratios

Fluid Properties	Haworth & SPE	GeochemLog (Fluid ID)	Mud Gas Ratios
	Very light dry gas	Methane (C1)	
	Light gas	Dry gas (DG)	
		Wet gas (WG)	
	Light oil	Gas condensate (GC)	
		Light/volatile oil (LO)	
	Medium gravity oil	Medium/black oil (MO)	
		Heavy oil (HO)	
	Residual oil	Residual oil/Bitumen (RB)	
	Coal bed gas		

C1 Content

The concentration of C1 in mud gas, expressed as percentage (%) or parts per million (ppm), or the ratio of C1 to total hydrocarbon gas (THC), is another widely used gas ratio parameter for fluid typing. The log of C1% typically illustrates the evolution of fluids with depth in the wells studied (Kandel et al., 2001). A higher C1 content generally indicates a lower gravity hydrocarbon fluid. Cutoff values can be established for distinguishing from dry gas to heavy oil using C1% or C1/THC (Ghneej et al., 2013). However, various factors such as lithology, changes in porosity, fluid contacts, and biodegradation may significantly affect C1 content in mud gas (Kandel et al., 2001). We recommend the use of C1 content for fluid typing in shale reservoir and as supplementary parameter in fluid typing for various formations.

Improved Haworth Method

A refined approach building upon the Haworth method has been developed, incorporating three key parameters: **Wh**, the differential between Wh and Bh (**Wh-Bh**), and C1/THC (**C1R**). This enhanced methodology enables the identification of eight distinct hydrocarbon fluid types and aligns well with established petroleum engineering standards (Table 1).

We have introduced five significant modifications to the conventional Haworth method:

- Using the fluid classification compatible to the criteria of reservoir and production engineering.
- Adding a cutoff of $Wh < 0.2$ (or $C1R > 98\%$ when Wh is not available) to indicate very dry gas (C1-dominant).
- Using Wh-Bh to quantitatively describe the separation of Wh and Bh and three cutoffs to help Wh distinguish between wet gas/condensate, light/medium oil, and heavy oil/residual oil.
- Unlike the fixed cutoff values used in the original Haworth method and its iterations, all cutoff values for Wh, Wh-Bh, and C1R are adjustable and can be calibrated for specific field applications. We have developed a set of cutoff values for mud gas fluid typing in the Paleozoic petroleum system of the Arabia Basin.
- We display Ch log with its cutoff (0.5) as a reference for fluid typing decision, rather than using Ch in the computer code for determining fluid types.

The original Haworth method (Haworth et al., 1985) proposed that "if $Ch < 0.5$, the Wh and Bh interpretation of gas is correct; if $Ch > 0.5$, the gas character indicated by the Wh and Bh ratios is associated with oil; and Ch is used only to clarify the interpretation of Wh and Bh when the indicate gas". Many subsequent studies used Ch in various ways, either as an auxiliary proxy to confirm gas or liquid phase

determined by Wh and Bh parameters (e.g., [Dolson, 2016](#)), or to differentiate gas, condensate and oil (e.g., [Mckinney et al., 2007](#)). Nevertheless, the underlying principle of Ch indication was not elucidated by [Haworth et al. \(1985\)](#) and [Whittaker \(1991\)](#), and there is no theoretical or statistical evidence to support the use of Ch for identifying condensate and confirming gas phase and gas association. In our experience, the use of Ch in fluid type determination can lead to confusion in fluid identification, as the simple Ch value may not adequately capture the complexity of gas association and condensation in mud gas logging and production

Application Examples

Multiple QC parameters, normalization of TGas/THC for hydrocarbon shows, and the improved Haworth method for fluid typing are integrated in a workflow and implemented in the software. This enables quick formation evaluation based on the cost-effective mud log data. More than 100 wells in basin scale are tested using the software. The fluid types inferred from mud gas ratio are systematically compared against the reservoir fluid samples collected from downhole, well head, and separator. Thermal maturity data and fluid prediction by basin modeling serve as supplementary fluid information for field calibration. To demonstrate the effectiveness of our approach, three examples are selected to showcase the fluid typing results and interpretation.

Well A: Fluid Typing for Paleozoic Petroleum System

Standard mud logging was conducted in well A that penetrated into Ordovician clastics, providing intact and continuous mud gas readings to cover over 11,000 ft of Paleozoic successions in the area ([Fig. 4](#)). The normalized THC (THC_n) in the track of HC Show demonstrates hydrocarbon shows in the shaly sand (SR), organic-rich shale (QB), and sand (SH) formations. These findings have been confirmed by well testing of this well and nearby wells. Gas wetness (Wh), which is a widely used proxy for thermal maturity gas geochemistry ([Jenden et al., 1993](#)), shows an increase maturity from a potential Devonian source rock (ST) to a Silurian proven source rock (QB). Fluid typing based on integration of Wh, Wh-Bh and C1R parameters indicates the transition from heavy oil at the top to very dry gas at the bottom of the log.

The vitrinite reflectance equivalent (VRE) based Rock-Eval and graptolite reflectance analyses on in the Devonian (VRE=0.6%) and Silurian (VRE>2.0%) source rock intervals, supports the maturity assessment and fluid typing based on mud gas data using our improved method. Two separator samples collected well testing confirmed the production of dry gas in the SH sandstone formation. The fluid typing results based on mud gas data are in good agreement with the source rock evaluation and fluid sampling, and provide additional details for the systematic change in thermal maturity and vertical variation of fluid types and properties within the well.

Well B: Fluid Typing for Multiple Pay Zones

The exploration well B was drilled to evaluate hydrocarbon potentials of porous intervals above, within, and below the Silurian organic-rich shale. Standard mud logging (SMG) was operated for the entire well, while advanced mud gas logging (AMG) was performed only for the target formations in the lower Paleozoic successions. The mud gas log provided the earliest and while drilling data for hydrocarbon shows and fluid typing, enabling the avoidance of missed pay and informing decisions regarding fluid and rock sampling, as well as optimizing well testing. Multiple pay zones were identified in the shaly sand (SR), shaly sand in middle of the organic-rich shale (QB), sand (SH), and tight sand (UW) formations in the Silurian and upper Ordovician clastic successions ([Fig. 5](#)). Our tool, utilizing mud gas data, predicted the presence of medium (black) oil in the pay zones within the SR and QB formations, which was subsequently confirmed by compositional PVT analyses of liquid samples (separator and downhole) from these intervals.

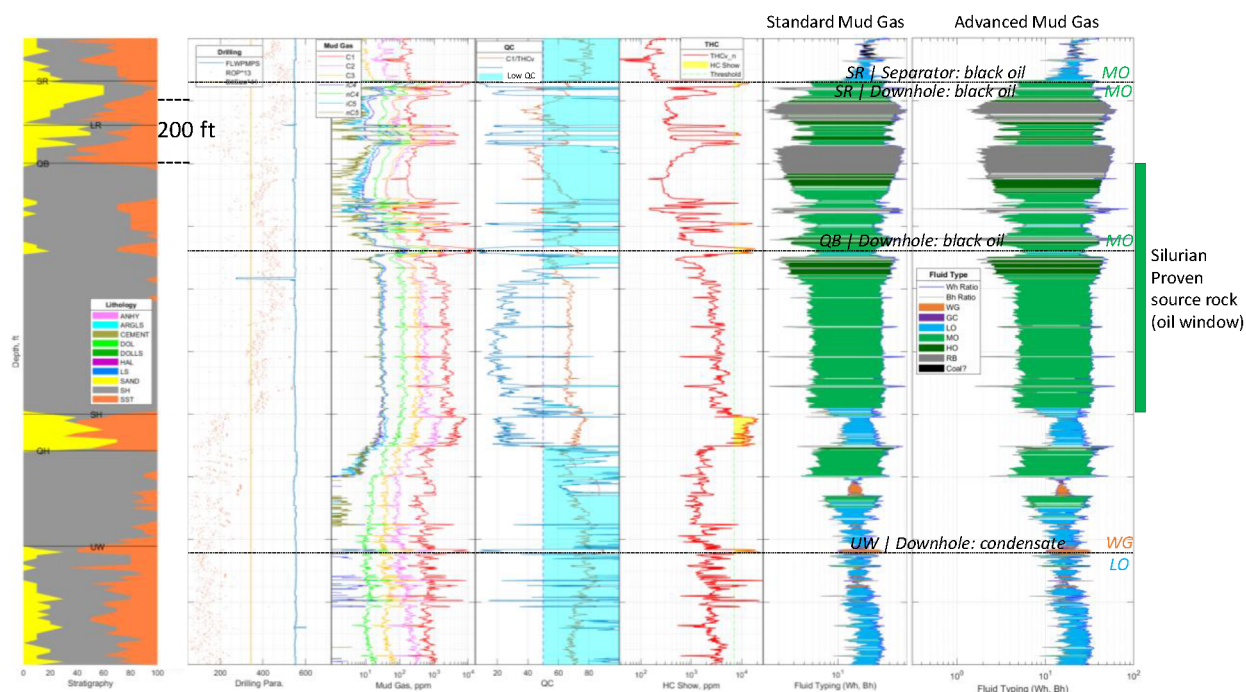


Figure 5—Well B: mud gas interpretation in GeochemLog demonstrates fluid typing and hydrocarbon show results in Silurian and upper Ordovician clastics. Multiple pay zones are identified in SR, QB, SH and UW formations. The last two tracks illustrate fluid typing results based C1-C5 data from standard mud gas and advanced mud gas data, respectively

Light (volatile) oil is predicted for the fluid trapped in the SH sand. Although there is no fluid sampling and well testing to confirm the prediction, the maturity of the source rock (QB) ($VRE = 0.86\%$, oil window) in the well, basin modeling, and nearby well testing results support the high possibility of light oil in the formation at the location. Wet gas was identified for the pay zone in the UW sand, with light oil present below, although the fluid sample was later clarified as condensate in PVT study. The mud gas ratio method has difficulty differentiating between wet gas, condensate and volatile oil. The fluid typing results based C1-C5 data from SMG and AMG show very similar results (Fig. 5). In our experience, AMG may capture more heavy gas components compared to SMG. Further comparison between AMG and SMG, as well as mud gas fluid typing and fluid sampling, is necessary to draw conclusions and determine the shift of mud gas ratio cutoffs for fluid typing.

Well C: Fluid Typing for Very Dry Gas

Well C was drilled in the center of the source rock kitchen, where the thermal maturity is extremely high ($VRE > 2.5\%$). The C1/THC ratio (C1R) exhibits very high values ($> 98\%$) at the bottom of the source rock (QB) formation, indicating that methane (C1) is dominant C1, if not the only, hydrocarbon component present in mud gas. In this scenario, the Wh and Bh values (see Equation 1 and 2) cannot be determined, which means that the original Haworth method and improved methods based on Wh and Bh ratios are unable to determine the fluid type, specifically very dry gas, as shown in Fig. 6.

To address the limitations, we have incorporated C1R as a supplementary parameter for the improved Haworth method for fluid typing, which can be used when either Wh or Bh is not available. This situation can occur not only in the case of overmatured thermogenic gas but also commonly with methane exsolution, coal-bed gas, and microbial gas (Golding et al., 2013). As shown in the last track of Fig. 6, the use of C1R as a supplementary parameter enables the identification of very dry gas (C1) and distinguishes it from normal dry gas, thereby providing a more accurate fluid typing result.

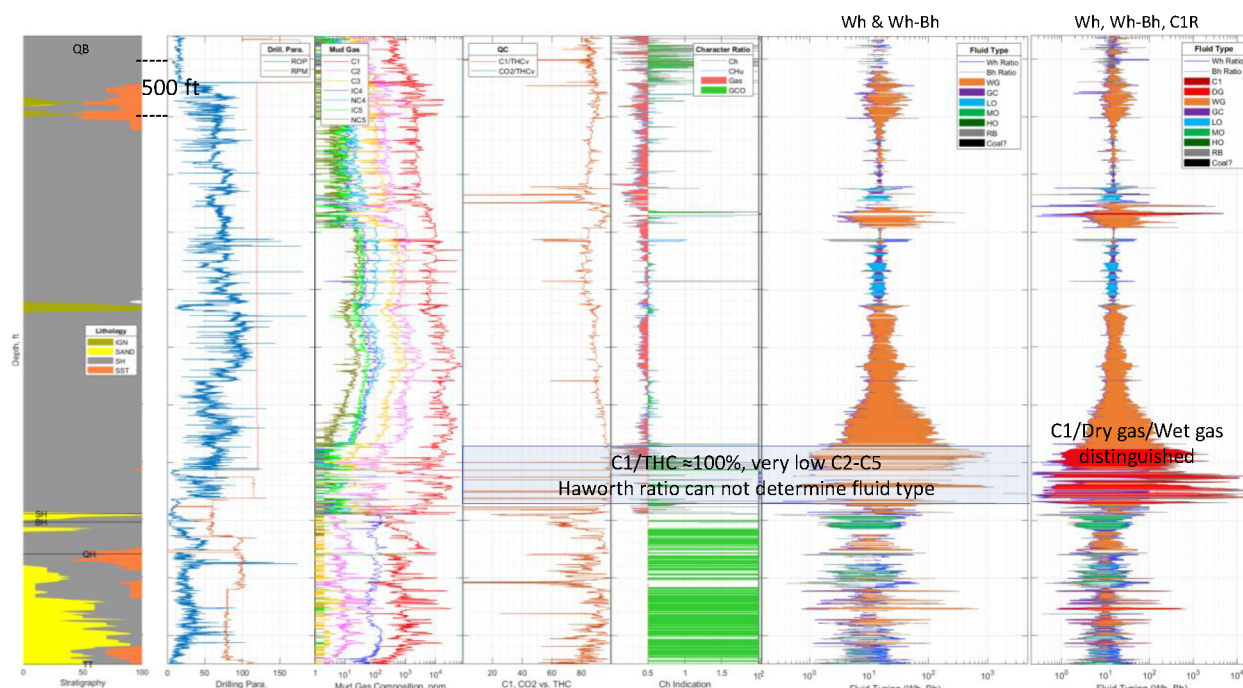


Figure 6—Incorporate C1R (C1/THC) as a supplementary parameter for fluid typing when Wh or Bh is not available?

Conclusions

In conclusion, this study presents a refined approach to mud gas fluid typing, leveraging three critical mud gas ratios (Wh, Wh-Bh, C1R) to identify eight distinct hydrocarbon fluid types in accordance with established petroleum engineering standards. The validation of this approach, integrated with mud gas quality control and total gas normalization in a computer program, provides a rapid and cost-effective formation evaluation tool for identifying hydrocarbon shows and fluids in real-time.

The application of this tool generates the first available information on pay zones and fluids for a well, supporting real-time decisions on fluid and rock sampling, formation and well testing, as well as subsequent wireline log interpretation and petroleum system analysis. However, it is noted that mud gas ratio methods can struggle to differentiate condensate from wet gas and volatile oil, highlighting the need for big fluid sample data, advanced mud gas logging, and machine learning techniques to accurately identify condensate and predict fluid properties in real-time.

The scalability of this technique enables its application to multiple wells within a field, facilitating the examination of vertical and lateral fluid variability and the mapping of migration pathways for mature field exploration and development. Ultimately, this refined approach and the tool have the potential to enhance the efficiency and effectiveness of exploration, formation evaluation and reservoir fluid characterization.

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